# ASK INSTEAD OF GUESS

## Handling Missing Information when AI Screens Patients for Clinical Trials

**MTP Progress Presentation**

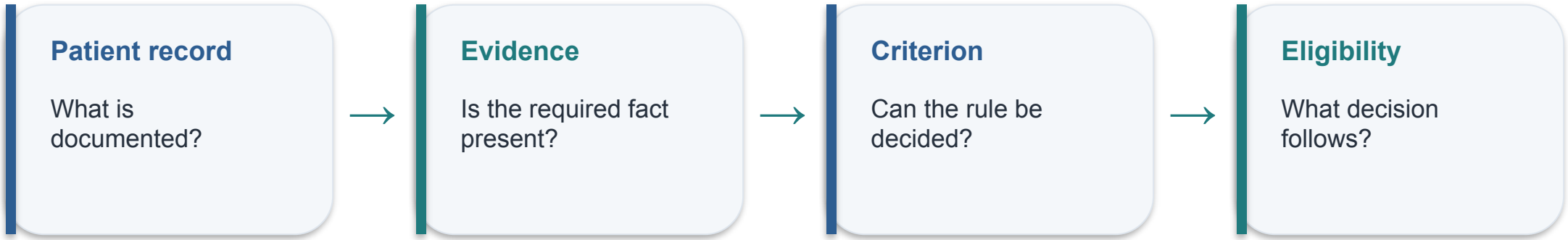
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Patient Eligibility for Clinical Trials

# Where clinical-trial screening can fail

**Eligibility screening is not only a prediction problem — it is also an evidence and measurement problem.**




**Missing evidence**

The record may omit the fact required by the criterion.

**Ambiguous / uncertain evidence**

A mention may not be sufficient to determine the rule.

**Reviewer or model interpretation**

The available text can still be interpreted incorrectly.

# Two prior annotation studies established the foundation

Related, but different targets — neither is a replication of the other.

### Historical NSCLC annotation audit

- 3 independent reviewer files
- 301 rows/file
- 251 unique items + 50 repeated items
- Labels: met / not_met / not_stated
- 300 aligned complete rows for 3-reviewer analysis

What it contributes:
Reviewer disagreement, within-reviewer inconsistency, and reviewer-dependent handling of missing evidence.

### Frozen SIGIR evidence-availability study

- 160 criterion-level items
- 28 patient–trial clusters
- 80 inclusion / 80 exclusion
- 2 blinded reviewers + adjudication
- Six evidence/determinability categories

What it contributes:
No evidence, partial evidence, uncertainty, unscorable representation, and the distinction between evidence presence and determinability.

# What the annotation metrics tell us

Use reliability metrics for annotation quality; use routing metrics only where a consensus target exists.

### Agreement

Observed agreement
How often labels match.

Cohen's κ / Fleiss' κ
Agreement beyond chance.

Duplicate consistency
Whether the same reviewer is stable on repeated items.

### Evidence availability

Any evidence
Does the record mention something relevant?

Determinability
Is there enough evidence to decide the criterion?

### Routing / decision metrics

Confusion matrix
Accuracy / balanced accuracy
Sensitivity / specificity
Precision / NPV / F1 / MCC

Plus workload, review, recovery and unresolved cases for Student 1's experiment.

**Key principle: evidence presence ≠ determinability.**

# Why `not_stated` is unstable

Historical NSCLC audit: reviewers applied the same three labels at noticeably different rates.

| Label | Reviewer A | Reviewer B | Reviewer C |
|---|---|---|---|
| **Met** | 103 | 83 | 96 |
| **Not met** | 87 | 76 | 45 |
| **Not stated** | 110 | 142 | 160 |
| **Labelled rows** | 300 | 301 | 301 |

### Three-class reliability

A/B: 0.660 agreement, κ = 0.484
A/C: 0.447, κ = 0.152
B/C: 0.502, κ = 0.201

### Overall consistency

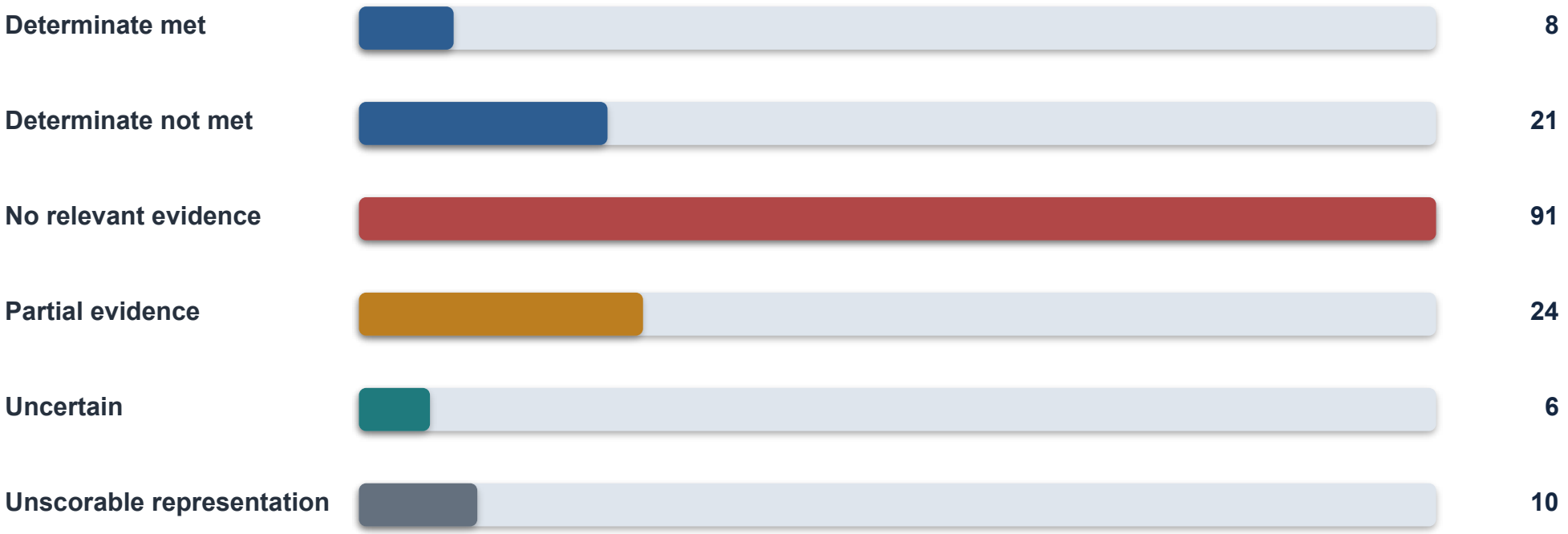
Fleiss' κ = 0.275
Only 103/300 had the same 3-class label across all reviewers.

### Duplicate consistency

A: 31/50 (0.62) | B: 47/50 (0.94) | C: 24/50 (0.48)
The use of `not_stated` was not stable across reviewers.

# Why missingness needs more than one label

Frozen SIGIR study: missing / unresolved evidence was separated into six categories.




### Evidence availability

53/144 had any evidence; 91/144 had none.

### Determinability

Only 29/144 were fully determinable; 115/144 were not fully determinable.

# Student 1's research question

## Can targeted questioning handle decision-relevant missing information more safely and efficiently than forced guessing or fixed defaults?

### When evidence is sufficient

Patient record
↓
Evidence-bound interpretation
↓
Criterion can be decided

### When evidence is missing

Patient record
↓
Preserve UNKNOWN
↓
Ask a decision-relevant question
↓
Re-decide only if the answer matters

**The experiment compares asking against guessing and fixed defaults — without assuming the outcome.**

# Student 1 annotation work: completed student labels

**These are student annotation assignments, not clinical ground truth.**

### NSCLC-100 — 446 rule answers

190 development + 256 test

Truth labels
- Yes: 111 (24.9%)
- No: 61 (13.7%)
- Unknown: 274 (61.4%)

Basis
- Stated: 163 (36.5%)
- Inferred: 12 (2.7%)
- Absent: 271 (60.8%)

Evidence span present: 175/446 (39.2%)
UNSURE:-prefixed comments: 0
Label-check: 1 conflict; supervisor disposition pending.

### SIGIR-204 — 174 rule answers

107 development + 67 test

Truth labels
- Yes: 23 (13.2%)
- No: 13 (7.5%)
- Unknown: 138 (79.3%)

Basis
- Stated: 31 (17.8%)
- Inferred: 5 (2.9%)
- Absent: 138 (79.3%)

Evidence span present: 36/174 (20.7%)
UNSURE:-prefixed comments: 5 (2.9%)
Label-check: 1 conflict; supervisor disposition pending.

# Evidence-bound Reader

Goal: convert an unstructured patient note into structured facts without inventing support.

### Patient note

Unstructured clinical text

### Reader

Structured patient facts
+ exact evidence spans

### Validation

Reject unsupported facts
Preserve uncertainty / negation

### Output log

Kept • rejected • unparseable
+ rejection reason

- Preserve negation, uncertainty and dates.
- Preserve comparators, numeric values and units.
- A fact is retained only when its supporting text is present in the note.

# Rule Decider: deterministic where possible

The decision layer separates clear structured rules from fuzzy textual judgment.

### Structured rule

Extracted fact
+
Deterministic rule

Examples: numeric ranges, presence/absence, comparators, units, synonyms

### Missing fact

Required fact not supported
→ UNKNOWN

Absence of evidence does not automatically mean the patient fails.

### Fuzzy text

AI textual judgment

Keep it separate from deterministic logic and preserve provenance.

**Report routing counts, corrections, error types, unit conflicts, synonym misses and engine tests.**

# Hidden-sentence experiment

Test whether the system can recover a decision when the deciding fact is deliberately hidden.

1

### Full note

Start from complete development notes.

2

### Find deciding sentence(s)

Identify the evidence that determines the rule.

3

### Hide

Remove 1 sentence or 2–3 fixed-seed sentences.

4

### Ask / decide

Run the selected policy with the hidden fact missing.

5

### Restore

Use the coordinator-held sentence as the answer source.

### Controls

Fixed seeds • unclear cases excluded • test cases built only after the relevant freeze • removed sentence retained as coordinator answer.

# Guess, defaults, unknown and ask

Controlled policy comparison on the same hidden-information cases.

| | | |
|---|---|---|
| **No questions**<br>Keep UNKNOWN | Review / unresolved | |
| **Forced guess**<br>Choose YES / NO | No unknown allowed | |
| **Lenient default**<br>Apply permissive default | Lower threshold to reject | |
| **Strict default**<br>Apply restrictive default | Higher threshold to reject | |
| **Targeted ask**<br>Ask a decision-relevant question | Recover information selectively | |

**Primary comparison question**

## Which policy recovers decisions while controlling wrongful inclusions, wrongful exclusions, review workload and unresolved cases?

It is not assumed that asking wins. The experiment is designed to determine whether targeted asking provides enough patient-level benefit for its cost.

# Question eligibility: ask only when the answer can matter

A question should be eligible for selection when it can resolve decision-relevant missing information.

| | | |
|---|---|---|
| 1 | **Decision-relevant** | The missing fact affects a criterion or downstream eligibility decision. |
| 2 | **Potentially changing** | A different answer could change the current decision. |
| 3 | **Answerable** | The requested information can plausibly be obtained from the available context. |
| 4 | **Worth the cost** | The expected decision recovery justifies the additional question/review workload. |

> **Core idea**
>
> Do not ask everything. Ask the missing fact most capable of changing the decision.

# Planned primary evaluation metrics

Evaluate at the patient–trial level as well as the rule level; always report counts with rates.

### Patient-level outcomes

Wrongful exclusions
Wrongful inclusions
Review cases
Decisions recovered by asking
Questions per case
Unresolved cases
No-information questions

### Decision metrics

Accuracy
Balanced accuracy
Sensitivity / recall
Specificity
Precision
NPV
F1
MCC

### Study controls

Full-note agreement
Question budget 0–6
Counts beside every rate
Cluster uncertainty by trial (NSCLC)
Cluster uncertainty by patient (SIGIR)
Prevalence-standardized planning at 10% and 5%

**Headline outcomes should reflect the decision: recovery and harm under a fixed workload.**

# Question-order strategies

Compare how different policies choose the next question under a fixed question budget.

| | |
|---|---|
| **Fixed list** | Predefined order |
| **Random** | Fixed-seed random order |
| **Development-set base rate** | Uses dev-only failure frequency |
| **AI-estimated failure probability** | Ranks by estimated failure probability |
| **Direct AI choice** | AI directly selects next question |
| **Oracle** | Upper-reference strategy using known answer |

**Budget** 0 → 1 → 2 → 3 → … → 6 questions

**Reference** Compare decision recovery against the full-note reference.

# NSCLC → SIGIR: keep the evaluations separate

Separate reporting is required because the datasets expose different failure surfaces.

| | NSCLC-100 | SIGIR-204 (Balanced dataset) |
|---|---|---|
| **Pairs** | 100 | 204 |
| **Trials** | 10 | 204 |
| **Patients** | 99 distinct | 14 |
| **Split** | 50 dev / 50 test<br>by trial | 102 dev / 102 test<br>by patient |
| **Rules** | 2,230 rule answers | 1,911 reviewed rules |
| **Error-analysis unit** | Trial | Patient |

**Do not pool the datasets or present a direct ranking between them.**

# Error analysis

When a decision fails, identify which layer failed before changing the policy.

### Template / routing

Wrong rule mapping or route

### Deterministic rule defect

Rule / comparator / unit logic wrong

### Reader omission / extraction

Required fact missed or unsupported

### Textual judgment

Fuzzy criterion interpreted incorrectly

### Synonym mismatch

Equivalent concept not recognized

### Student-label error

Annotation later corrected or flagged

**Report separately: NSCLC by trial • SIGIR by patient • never pool the two error surfaces.**

# Current status and next milestones

Completed annotation work is separated from experiments that have not yet been run.

| | |
|---|---|
| Completed | 301-row annotation exercise |
| Completed | 160/160 reviewer task |
| Completed | 446 NSCLC student annotation pass |
| Completed | 174 SIGIR student annotation pass |
| Completed / review pending | NSCLC label-check: 1 conflict |
| Completed / review pending | SIGIR label-check: 1 conflict |
| Next | **Freeze annotation files + record hash / dispositions** |
| Next | **Measure shared-row agreement + Cohen's κ** |
| Next | **Run Reader / Rule Decider experiments** |
| Next | **Run hidden-sentence + guess/default/ask comparisons** |
| Next | **Evaluate question ordering and patient-level outcomes** |

**No empirical system-performance claims are made in this progress deck.**

# Limits and possible negative findings

The experiment is designed to allow a negative result.

- Questions may fail because the missing information is not obtainable from the accessible record.
- Questioning may add too much review / interaction workload.
- Simple defaults may perform similarly to more complex policies on some cases.
- A simple question order may match an AI-based order.
- Reader extraction may become the bottleneck rather than question selection.
- Some unresolved criteria may remain unresolved even after asking.

**Decision rule adopted**

Targeted asking is useful only if it reduces wrongful exclusions or improves full-note decision recovery without unacceptable increases in wrongful inclusion, unanswered questions or workload.

**If the answer is no, the negative result is still a valid research result.**

# Check What Can Change the Outcome

Targeted evidence checking for AI-based clinical-trial screening

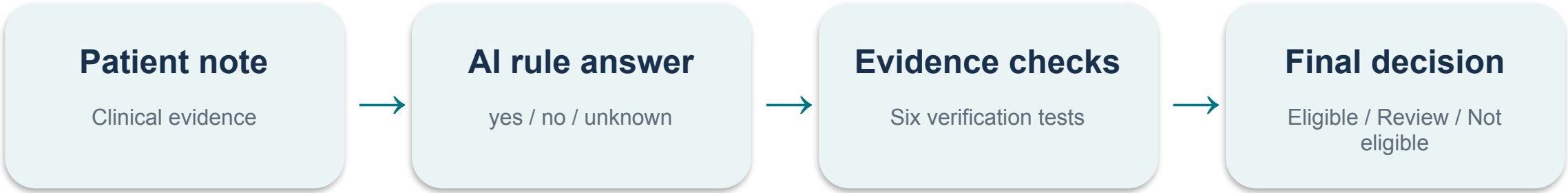
Technical validation, prior evidence, proposed methods and next steps

# Why checking matters

- An AI model answers trial eligibility rules using a patient note.
- A plausible citation does not guarantee that the answer is supported.
- A single incorrect rejection reason may exclude an eligible patient.
- Our focus: check the answers most likely to change the final decision.

# Screening workflow and where checking helps

Conceptual process — proposed verification component




**Prioritize errors that could change eligibility**

# Two related research directions

## Ask Instead of Guess — Aman

Identify missing information.

Keep UNKNOWN when evidence is insufficient.

Ask focused follow-up questions.

## Check What Can Change — Akash

Verify AI rule answers.

Identify evidence and reasoning errors.

Prioritize checks by decision impact.

Both projects use the NSCLC and SIGIR clinical-trial screening resources.

# Why a checker is needed

- Historical NSCLC audit: 77 of 151 model-cited violations were genuine (50.99%).
- Historical human–model study: 11 joint errors versus 5.3 expected (2.08×).
- Frozen SIGIR study: 95 of 144 items routed for review (65.97% workload).
- These findings are prior laboratory evidence, not results from our new checker.

# Why targeted verification matters

Prior laboratory findings — not results of the proposed student checker




**Human–model errors: 11 observed together vs 5.3 expected (2.08×)**

# Two datasets and their evaluation splits

## NSCLC-100

100 patient–trial pairs

10 trials; 223 reviewed rules

2,230 criterion answers

50 development / 50 test

Split by trial

## SIGIR-204

204 patient–trial pairs

14 patients; 1,911 reviewed rules

100 Eligible / 104 Not Eligible

102 development / 102 test

Split by patient

Do not tune on the held-out test splits.

# Technical validation completed

- NSCLC starter-kit self-tests: 36 passed, 0 failed.
- SIGIR package checks: 9 passed, 0 failed.
- Python environment verified with NumPy 1.26.4 and Pandas 2.2.3.
- Qwen2.5:7b downloaded and responded successfully via local Ollama.
- Real clinical-trial inference was started; no completed model accuracy result yet.

# Baseline and evaluation pipeline

- Plain AI: one local-model answer per patient–trial rule.
- Each response contains yes / no / unknown and supporting quotation.
- Rule polarity maps answers to met / not_met / not_stated.
- Pair decision: any failed rule → NOT_ELIGIBLE; all met → ELIGIBLE; otherwise REVIEW.
- Compare predictions with validated annotations and pair-level labels.

# Six planned evidence checks

- 1. Citation existence — is the quoted span in the note?
- 2. Support — does the quote justify the claimed answer?
- 3. Context — negation, family history, past or hypothetical events.
- 4. Logic — rule polarity and numerical thresholds.
- 5. Contradiction — conflicting evidence elsewhere in the note.
- 6. Missed evidence — decisive information ignored by UNKNOWN.

# Why decision impact determines priority

## Single rejection reason

One failed criterion is the only reason for rejection.

If it is incorrect, correction could change the pair decision.

## Multiple rejection reasons

Several failed criteria remain.

Fixing just one may not change the pair decision.

Decision impact is calculated from rule outcomes, not model confidence alone.

# Compare checking strategies at equal budgets

- Random selection — fixed-seed reference.
- Low confidence — inspect uncertain answers first.
- Checker flags — inspect suspected evidence errors first.
- Decision impact — inspect pivotal rejection reasons first.
- Combined strategy — evidence warnings plus decision impact.
- Evaluate at 5, 10, 20, 50 and 100 checks.

# Decision impact: a worked example

Illustrative rule logic, not a measured patient result

## One failed rule

Rule A: MET

Rule B: NOT MET

Rule C: MET

**Fix B → ELIGIBLE**

## Two failed rules

Rule A: NOT MET

Rule B: NOT MET

Rule C: MET

**Fix A → still NOT ELIGIBLE**

# Traps, repair and deferral

- Controlled traps: negation, family history, hypothetical statements, old values, contradictions, removed evidence and threshold changes.
- Compare three actions: keep the answer, defer to UNKNOWN/review, or re-ask once with feedback.
- Measure both incorrect answers fixed and correct answers broken.
- Main proposed outcome: eligible patient–trial pairs rescued per 10 checks.

# Completed work and remaining work

## Completed / verified

- Both datasets and splits validated
- 36 NSCLC tests passed
- 9 SIGIR checks passed
- Qwen2.5:7b responds locally
- Baseline script and evaluation CLI available

## In progress / not yet measured

- Independent truth annotations
- Real baseline accuracy
- Six-checker implementation
- Priority-strategy comparison
- Repair and trap experiments

No mock-model output is presented as real clinical-model performance.

# Immediate development plan

- Complete and freeze a small independent development annotation subset.
- Run and evaluate a bounded Qwen2.5:7b development pilot.
- Implement evidence checks and inspect errors manually.
- Compare review priorities on the same checked examples.
- Freeze the method before transferring it to SIGIR and the held-out test sets.

# Current implementation status

Verified technical work versus experiments still to complete

| | |
|---|---|
| NSCLC integrity and unit tests | 36 / 36 PASS |
| SIGIR integrity checks | 9 / 9 PASS |
| Local Qwen2.5:7b model | READY response |
| Real model baseline metrics | Not measured |
| Six-check evidence checker | Not implemented |
| Priority and repair comparison | Not implemented |

# Limitations and interpretation

- A small pilot will not establish generalization to all trials.
- Incomplete notes make some criteria genuinely indeterminable.
- Review workload matters as much as raw checker recall.
- Automated repair may introduce new errors; deferral may be safer.
- Report negative findings and incomplete experiments explicitly.

# PRELIMINARY EXPERIMENTAL RESULTS

Qwen2.5:7b baseline | NSCLC-100, pair 7 (T01) | exploratory n = 1 pair

**21 / 21** — Rule answers written

**0** — Failed model calls

**1 / 1** — Eligible-labeled pair predicted NOT_ELIGIBLE

**8 / 9** — Quoted spans matched verbatim

## Model rule-verdict distribution (21 criteria)



Met: 5 (23.8%) Not met: 2 (9.5%) Not stated: 14 (66.7%)

**Official evaluator: strict accuracy 0/1; this is NOT an overall model accuracy estimate.**

# PILOT CASE STUDY: WHY CHECKING MATTERS

Manual inspection of model output; automated six-check system not yet evaluated

| | | |
|---|---|---|
| **Logic: ECOG 0 or 1** | Patient note: ECOG=1; model verdict: NOT MET | **Confirmed model logic error** |
| **Citation existence: NSCLC** | Model quoted criterion text absent from patient note | **Non-verbatim citation** |
| **Tumor size / reference label** | Stage Ia, tumor ≤2 cm; trial requires >2 cm | **Model rejection appears justified; label requires review** |

**Interpretation: correcting ECOG alone does not rescue the pair; another criterion still fails.**

Scope: one-pair pilot. Checker detection rates, repair outcomes, and rescue-per-check remain unmeasured.

# Thank you

## Check what can change the outcome.

Questions and discussion